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# Remagnetization of Jurassic volcanic rocks in the Santa Rita and Patagonia Mountains, Arizona: Implications for North American apparent polar wander

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**Abstract.** Paleomagnetic poles for the Jurassic Corral Canyon sequence and Glance Conglomerate in southern Arizona have been used to construct apparent polar wander (APW) paths for the North American plate, but they are controversial and conflict with higher-latitude poles from New England. Lower Jurassic dacites and ash flow tuffs of the Mount Wrightson Formation in the Santa Rita Mountains were initially sampled to provide an additional paleopole for southern Arizona. These rocks, however, have a predominantly reversed-polarity characteristic magnetization (in situ, $I = -47°$, $D = 154°$, $\alpha_{95} = 9°$) which is statistically indistinguishable from that for the nearby latest Cretaceous Elephant Head pluton ($I = -48°$, $D = 165°$, $\alpha_{95} = 8°$). Although magnetizations of both polarities are observed in the ash flow tuffs, they are mostly carried by hematite, and dual polarity components are observed within some specimens. Moreover, widespread mineralization and a K-Ar age of ~67 Ma for altered rocks of the Mount Wrightson Formation imply that these rocks were subjected to a prolonged episode (greater than one polarity interval) of low-temperature alteration and remagnetization. Middle Jurassic ash flow tuffs of the Corral Canyon sequence in the Patagonia Mountains to the south have a similar geologic setting: They are exposed along strike amid mineralized rocks on the eastern flank of a mountain range cored by Paleocene intrusions. Hematite is also the dominant remanence carrier in most of the Corral Canyon sequence, and its predominantly normal-polarity direction (in situ, $I = 51°$, $D = 326°$, $\alpha_{95} = 9°$) is indistinguishable from that for the nearby Patagonia Granodiorite ($I = 49°$, $D = 342°$, $\alpha_{95} = 8°$). Rocks of the Corral Canyon sequence therefore are likely remagnetized as well. Problems also exist with the Glance Conglomerate pole. These rocks are situated within a caldera structure and have been potassium metasomatized. This potassic alteration could have occurred shortly after emplacement or at a later time, postdeformation. The low-latitude Jurassic APW path for North America and J-2 cusp therefore are not well supported and may need revision.

## Introduction

Jurassic volcanic rocks in southern Arizona have been the subject of previous paleomagnetic investigations to determine cratonic reference poles and the apparent polar wander (APW) path for the North American plate. Assemblages of Upper Jurassic silicic tuffs and flows, including interbedded volcaniclastic conglomerates and sandstones, have been sampled in the Glance Conglomerate of the northern Canelo Hills [*Kluth et al.*, 1982], and similar rocks of Middle Jurassic age have been sampled in Corral Canyon of the Patagonia Mountains [*May et al.*, 1986]. A conglomerate test (Canelo Hills), normal- and reversed-polarity magnetizations, and accompanying isotopic ages imply that these rocks are reliable recorders of the Jurassic geomagnetic field direction for southern Arizona. *May and Butler* [1986] combined their proposed Jurassic poles and other reliable poles for stable North America and remodeled the Mesozoic APW path using paleomagnetic Euler pole (PEP) analysis [*Gordon et al.*, 1984]. They proposed an arcuate APW path that remains at low latitudes (near 60°N latitude) during the Middle Jurassic and changes direction sharply in Late Jurassic time forming the J-2 cusp (Figure 1).

More recently published paleomagnetic data from the Moat Volcanics in New Hampshire [*Van Fossen and Kent*, 1990], however, have been interpreted as indicating high-latitude APW (near 75°N latitude) during the Middle Jurassic, generally supporting the configuration of *Irving and Irving*'s [1982] "running-mean" path (Figure 1). In addition, secondary magnetizations for Upper Triassic and Lower Jurassic sedimentary rocks of the Newark basin [*Witte and Kent*, 1989, 1990], and secondary magnetizations for igneous rocks of the Newark Supergroup [*Smith and Noltimier*, 1979; *Witte et al.*, 1991], corroborate Middle Jurassic APW at high latitudes. Disagreement between low-latitude Middle Jurassic (~60°N) poles from Arizona and higher-latitude Middle Jurassic (~75°N) poles from New England therefore implies that problems exist with either or both of these data sets.

One problem in southern Arizona is the potential for remagnetization of Jurassic rocks by younger igneous activity. *Van Fossen and Kent* [1990] suggest that the in situ Corral Canyon results are based on a Cretaceous remagne-

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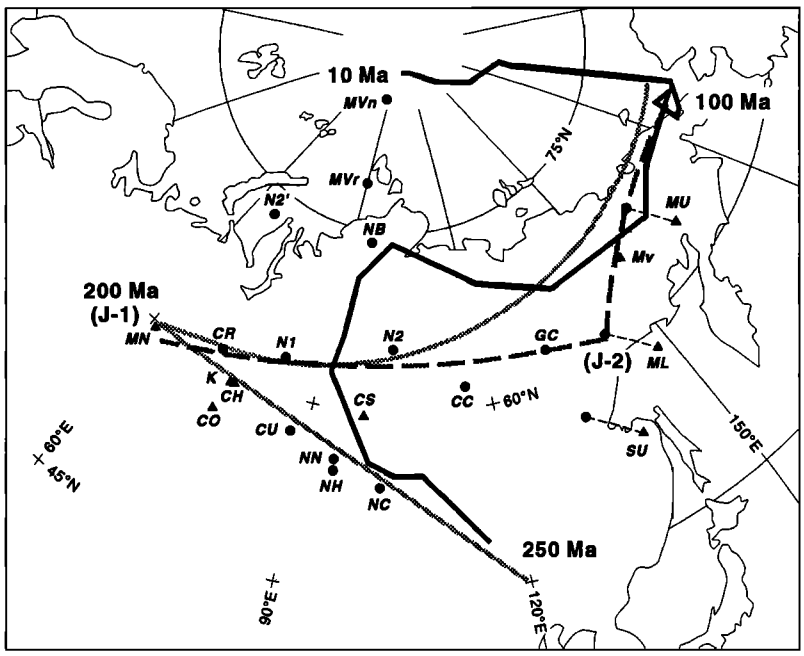



**Figure 1.** Mesozoic paleomagnetic poles and apparent polar wander paths for North America. APW paths are solid line, *Irving and Irving* [1982]; solid shaded line, *Gordon et al.* [1984]; dashed shaded line, *May and Butler* [1986]. Poles are (triangles indicate Colorado Plateau poles corrected for 5° of clockwise rotation around pivot at 37°N, 257°E) NC, NN, NH, Newark Basin Carnian-Norian, Norian, and Hettangian strata, respectively [*Witte et al.*, 1991]; Newark Supergroup N1 and N2 [*Smith and Noltimier*, 1979]; Chinle Formation, CS, Shinarump Member (Carnian); CU, upper shale member (lower Norian); CO, Owl Rock Member (middle Norian); CR, Redonda Member (middle and upper Norian); CH, Church Rock Member (middle and upper Norian); K, Kayenta Formation (Pliensbachian) [*Bazard and Butler*, 1991]; MN, Sinemurian Moenave Formation [*Ekstrand and Butler*, 1989]; MVn, MVr, ~166 Ma Moat Volcanics, normal and reversed polarities, respectively [*Van Fossen and Kent*, 1990]; NB, Newark Basin B component (~175 Ma); N2′, in situ Newark Supergroup N2 [*Witte and Kent*, 1991]; CC, ~172 Ma volcanic rocks of Corral Canyon [*May and Butler*, 1986]; GC, ~151 Ma Glance Conglomerate [*Kluth et al.*, 1982]; SU, upper Callovian(?) Summerville Formation [*Bazard and Butler*, 1992]; Morrison Formation, ML and MU, lower and upper parts, respectively [*Steiner and Helsley*, 1975]; Mv [*Van Fossen and Kent*, 1992b].

tization. Alternatively, Mesozoic and younger tectonic deformation in southern Arizona may have affected the rocks used to determine Jurassic pole positions [*Van Fossen and Kent*, 1992d]. Paleomagnetic investigations of Jurassic and Cretaceous rocks indicate significant clockwise vertical axis rotations (~30°–60°) in southern Arizona relative to cratonic North America [*Hagstrum and Sawyer*, 1989; *Hagstrum and Lipman*, 1991a, b], but removing such rotations cannot bring these southwestern poles into agreement with those from New England. Similar problems of remagnetization and unrecognized tectonism may also apply to the New England rocks. The Middle Jurassic Moat Volcanics pole from the Jurassic and Cretaceous White Mountains Magma Series [*Van Fossen and Kent*, 1990] is based on a magnetization which postdates tilting; the age of this pole has not been reliably determined [*Butler et al.*, 1992]. Moreover, the observed magnetization for the Moat Volcanics could be a Cretaceous overprint that has been tilted ~25° to the west [*Hagstrum*, 1993].

This study of the relatively unmetamorphosed Mount Wrightson Formation in the Santa Rita Mountains (Figure 2) was initially undertaken to determine an additional Jurassic paleopole for southern Arizona. Assessment of unpublished paleomagnetic data for the latest Cretaceous Elephant Head Quartz Monzonite and Paleocene Gringo Gulch pluton in the Santa Rita Mountains (R. F. Butler, written communication, 1992), however, implies that the Mount Wrightson Formation was remagnetized in the Late Cretaceous to early Tertiary. The objectives of this investigation, therefore, were changed, and an additional collection of samples was obtained from the Paleocene Patagonia Granodiorite in the Patagonia Mountains to determine if the nearby Corral Canyon sequence was also remagnetized by Late Cretaceous to early Tertiary plutonism.

## Regional Geology

Subduction-related volcanic and plutonic activity began in southwest North America during Late Triassic or Early Jurassic time [*Dickinson*, 1981]. The Late Triassic to Middle Jurassic magmatic arc in southern Arizona was established across a Paleozoic cratonic platform representing stable margin sedimentation, and these Paleozoic rocks overlie Precambrian (1.8–1.4 Ga) basement rocks [*Silver*, 1978]. The Lower Jurassic Mount Wrightson Formation is among the oldest preserved arc-related sequences in southern Arizona and represents a multivent volcanic complex deposited in a subsiding intra-arc basin [*Riggs and Busby-Spera*, 1990].

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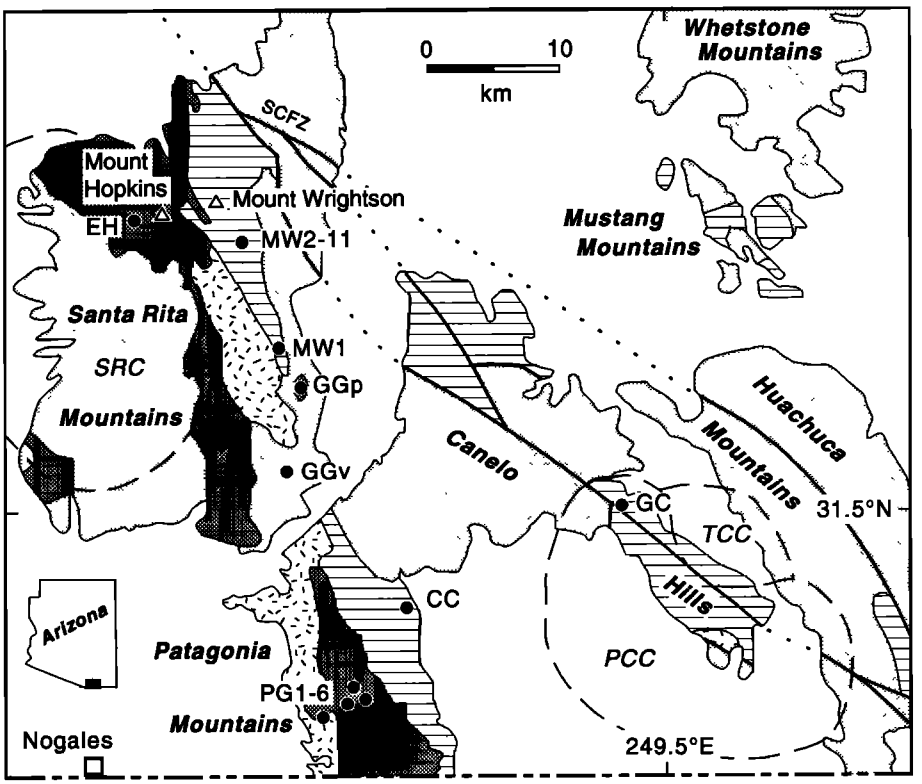



**Figure 2.** Geologic map of study area in southern Arizona showing paleomagnetic sampling locations in the Mount Wrightson Formation (MW1–11) and Patagonia Granodiorite (PG1–6, Table 1). Patterned areas are Jurassic volcanic rocks (horizontal lines), intrusive rocks (random line segments), and Late Cretaceous and early Tertiary intrusive rocks (dark gray). Heavy lines are faults belonging to the Sawmill Canyon fault zone (SCFZ). Dashed lines denote inferred caldera margins: SRC, Cretaceous Santa Rita caldera; PCC and TCC, Jurassic Parker Canyon and Turkey Creek calderas. Other paleomagnetic sampling sites are GC, Glance Conglomerate; CC, Corral Canyon sequence; GGp, Gringo Gulch pluton; GGv, Gringo Gulch Volcanics; and EH, Elephant Head Quartz Monzonite.

The Mount Wrightson Formation is a homoclinally dipping (~20°–35°NE) section of volcanic, hypabyssal intrusive, and lesser sedimentary rocks exposed along a 25-km strike length on the southeastern flank of the Santa Rita Mountains (Figure 2). Presently about 5 km thick, the Mount Wrightson Formation has been divided into lower, middle, and upper members [*Drewes*, 1971]. The lower member (~220 Ma U-Pb zircon age [*Asmerom et al.*, 1990]) consists predominantly of dacite and andesite lava flows intruded by penecontemporaneous andesite, latite, and dacite bodies. The middle member (206 ± 9 Ma, U-Pb zircon [*Riggs et al.*, 1986]) consists predominantly of dacite to rhyolite ash flow tuffs and eolian sandstone intruded by intermediate to silicic rocks. The upper member consists of eolian sandstone and andesite to dacite intrusions [*Riggs and Busby-Spera*, 1990].

The Mount Wrightson Formation is intruded along its base by Early Jurassic through Late Cretaceous and early Tertiary plutons. Among these plutons are the Elephant Head Quartz Monzonite and the Josephine Canyon Diorite (~69–71 Ma K-Ar biotite ages [*Drewes*, 1971; *Reynolds et al.*, 1986]) in the southwestern Santa Rita Mountains. These rocks have been interpreted as ring intrusions on the north and east sides of the Santa Rita caldera (Figure 2) from which the Upper Cretaceous ash flow tuff of the Salero Formation was erupted [*Lipman and Sawyer*, 1985]. Intruding into Lower Cretaceous strata to the east is the Gringo Gulch pluton (~69 Ma K-Ar hornblende age [*Reynolds et al.*, 1986]) which is composed of dacite porphyry and microgranodiorite phases.

Metamorphism of the Mount Wrightson Formation is generally subgreenschist grade and is confined to within 50 m of intrusion contacts. Preservation of volcanic and sedimentary textures is usually excellent, but hydrothermal alteration is also common [*Riggs and Busby-Spera*, 1990]. Widespread mineralization in the region and a K-Ar sericite age of 67 ± 2 Ma on a quartz-sericite-pyrite alteration zone in volcanic rocks [*Shafiqullah et al.*, 1980] attest to the prevalence of Late Cretaceous to early Tertiary alteration within the Mount Wrightson Formation. Late Jurassic through early Tertiary tectonism has formed the structural boundaries of the Mount Wrightson Formation but has affected it little internally. The Mount Wrightson Formation is truncated to the northeast by the northwest trending Sawmill Canyon fault zone (Figure 2), which may have been episodically active since Precambrian time [*Davis*, 1981]. Northwest to northeast trending, high-angle faults of uncertain age also cut the Mount Wrightson Formation but apparently cause no repetition of section [*Riggs and Busby-Spera*, 1990].

The Middle Jurassic Corral Canyon sequence in the northeastern Patagonia Mountains is at least 650 m thick and is composed of red volcanic conglomerate, sandstone, and

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siltstone; red to gray latitic or dacitic partially welded to densely welded tuff; and minor amounts of rhyolitic to dacitic lava flows [*Simons*, 1972, 1974]. To the north of the Corral Canyon fault these rocks generally form a homoclinal section, although dip angles range from 26° to 68° NNE; a Rb-Sr isochron for eight whole rock samples of welded tuff gives an age of 172 ± 6 Ma [*May et al.*, 1986]. The core of the Patagonia Mountains is the large body of medium- to coarse-grained Patagonia Granodiorite. K-Ar ages for these rocks are 60 ± 3 Ma (biotite), 60 ± 5 Ma (hornblende) [*Marvin et al.*, 1973; *Reynolds et al.*, 1986], and 65 ± 2 Ma (biotite) [*Damon et al.*, 1964; *Reynolds et al.*, 1986]. Along the northeast contact of the granodiorite, Mesozoic volcanic and sedimentary rocks have been hornfelsed, and Paleozoic carbonate rocks have been metamorphosed to garnetite, tactite, and marble. In addition, hydrothermal alteration and mineralization of middle Paleozoic to early Tertiary rocks occur at many places in the Patagonia Mountains. Hydrothermal circulation apparently ceased by middle Tertiary time because no alteration or mineralization is evident in rocks of Oligocene or younger age [*Simons*, 1974].

## Paleomagnetism

Samples for paleomagnetic analysis were collected at 11 sites in the lower and middle members of the Mount Wrightson Formation (Figure 2). Site MW1 is in dacite lava flows toward the base of the formation near the Dragon Z mine, and sites MW2-MW11 (taken progressively downsection) are in individual ash flow tuffs of the middle member exposed within a continuous stream wash exposure at the head of Walker basin. Unpublished paleomagnetic data for five sites in roadcut exposures of Late Cretaceous Elephant Head Quartz Monzonite on Mount Hopkins and for 16 sites in the Paleocene Gringo Gulch pluton were provided by R. F. Butler (University of Arizona) (Table 1). Drill core samples of the Paleocene Patagonia Granodiorite in the Patagonia Mountains were also collected at six sites from stream wash exposures in Sycamore and Providencia canyons.

From 3 to 10 core samples were taken at each site in the Mount Wrightson Formation (six to eight in the Patagonia Granodiorite) and were oriented using both magnetic and solar compasses. Attitudes of the tuff units were measured from eutaxitic structures or interbedded sedimentary rocks and, due to their similarity, were averaged to make a single overall correction (326°, 26°NE). This correction was also applied to data for the dacite flows (MW1) in which no paleohorizontal indicator was observed. Structural corrections were made by simple rotation of bedding to horizontal about the strike axis.

The Mount Wrightson Formation specimens were subjected to progressive thermal demagnetization to ~700°C, and remanent magnetizations were measured with a cryogenic magnetometer. Specimens of the plutonic rocks were demagnetized using alternating fields with peak inductions between 1 and 100 mT at the University of Arizona (EH1–5 and GG1–16) and U.S. Geological Survey (PG1–6). Least squares lines were fitted to the demagnetization data using standard procedures [*Kirschvink*, 1980], and site and overall directional means were calculated using *Fisher* [1953] statistics. Acquisition of isothermal remanent magnetization (IRM) experiments in fields up to 700 mT were made on selected specimens from each sampling locality of the Mount Wrightson Formation to help identify their magnetic mineralogy.

Natural remanent magnetizations (NRMs) for sites in the Mount Wrightson Formation were highly variable in intensity, direction, and degree of within-site dispersion, and NRM directions for the Patagonia Granodiorite samples were also moderately to highly dispersed. Demagnetization behavior for the Mount Wrightson Formation rocks ranges from being simple to complex. Univectorial normal-polarity magnetizations were observed in the dacite flows (Figure 3a) and in densely welded tuffs (MW9–11) at the bottom of the sampled section (Figure 3d). Multiple components of normal- and reversed-polarity magnetizations were observed in individual specimens (Figure 3b) from sites MW5, MW6, and MW8, and samples with primarily single components of either reversed- or normal-polarity magnetization were taken from these sites and from sites MW2 and MW3. The means of the reversed- and normal-polarity site-mean directions (Table 1), compared by inverting the reversed-polarity direction through the origin, are statistically indistinguishable from each other at the 95% confidence level [*McFadden and Lowes*, 1981]. Thermal demagnetization and IRM curves for samples of the Mount Wrightson Formation indicate that the magnetic remanence is carried by either magnetite (Figure 4, curve 1) or hematite (Figure 4, curve 4), or both (Figure 4, curves 2 and 3). Samples from most sites are dominated by hematite and have IRM curves similar to Figure 4, curve 3 or curve 4. In the Mount Wrightson Formation samples there is no apparent relationship between remanence polarity and magnetization carrier as indicated by demagnetization temperatures.

After removal of a viscous component with a direction near that of the present-day field, the intrusions generally showed a single component of remanent magnetization (Figure 5a). Samples from three sites in the Patagonia Granodiorite (PG4 and PG6) showed the presence of an additional higher-coercivity, reversed-polarity component of magnetization (Figure 5b). Directions for these sites are streaked between the observed normal-polarity direction and steep upward directions to the southeast, but converging remagnetization circles were not clearly defined by the demagnetization data. In situ site- and unit-mean directions (Table 1) for the Elephant Head Quartz Monzonite ($I = -48.2°$, $D = 164.9°$, $N = 5$, $\alpha_{95} = 8.2°$) and Gringo Gulch pluton ($I = -61.9°$, $D = 151.2°$, $N = 16$, $\alpha_{95} = 3.6°$) are shown in Figure 6a. Both have reversed-polarity characteristic magnetizations with directions eastward of the antipode of the expected North American Paleocene direction for southern Arizona ($I = 55.5°$, $D = 351.2°$) [*Diehl et al.*, 1983]. The difference in direction between the Elephant Head and Gringo Gulch plutons could be due to either differential tilting or to paleosecular variation of the geomagnetic field between the time of cooling and magnetization acquisition for these two plutons.

Figure 6b shows site- and unit-mean directions for the Mount Wrightson Formation in geographic coordinates (uncorrected for tilt). Normal-polarity directions for sites MW1, MW5, MW6, and MW8 have been inverted through the origin for comparison with the other reversed polarity site-mean directions. The unit-mean directions for the Elephant Head Quartz Monzonite, Gringo Gulch pluton, and Gringo Gulch Volcanics [*Vugteveen et al.*, 1981] are also shown,

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**Table 1.** Paleomagnetic Data for the Santa Rita Mountains and Patagonia Mountains, Southeastern Arizona

| Sites | $I$, deg | $D$, deg | $N$ | $R$ | $k$ | $\alpha_{95}$, deg | $\lambda$, deg | $\phi$, deg |
|---|---|---|---|---|---|---|---|---|
| | | | *Mount Wrightson Formation (31.7°N, 249.2°E)* | | | | | |
| MW1 | 48.4 | 325.6 | 8 | 7.9222 | 90 | 5.9 | 60.4 | 163.8 |
| MW2 | −52.1 | 154.4 | 9 | 8.7399 | 31 | 9.4 | 68.4 | 168.7 |
| MW3 | −51.1 | 141.7 | 5 | 4.9721 | 143 | 6.4 | 57.6 | 169.7 |
| MW4 | −52.5 | 133.1 | 7 | 6.3612 | 9 | 20.8 | 50.7 | 174.2 |
| MW5 | 41.9 | 354.6 | 8 | 7.9756 | 286 | 3.3 | 81.0 | 102.9 |
| MW6 | 39.6 | 4.4 | 5 | 4.9196 | 50 | 11.0 | 80.0 | 45.0 |
| | −53.3 | 153.5 | 6 | 5.9512 | 102 | 6.7 | 67.6 | 172.0 |
| MW7 | −32.8 | 157.1 | 3 | 2.9033 | 21 | 27.8 | 65.1 | 130.9 |
| MW8 | 39.3 | 312.9 | 3 | 2.9186 | 25 | 25.4 | 47.3 | 158.5 |
| Average | −47.3 | 153.7 | (9) | 8.7594 | 33 | 9.1 | 67.1 | 157.8 |
| Average/c | −38.1 | 176.7 | (9) | 8.7594 | 33 | 9.1 | 79.3 | 85.8 |
| MW9 | 84.9 | 19.2 | 5 | 4.9960 | 1005 | 2.4 | 41.2 | 253.6 |
| MW10 | 84.8 | 76.4 | 10 | 9.9738 | 343 | 2.6 | 33.6 | 261.3 |
| MW11 | 88.5 | 122.2 | 7 | 6.9887 | 532 | 2.6 | 30.1 | 252.1 |
| Average | 86.8 | 56.8 | (3) | 2.9962 | 527 | 5.4 | 35.0 | 255.7 |
| Average/c | 60.8 | 56.1 | (3) | 2.9962 | 527 | 5.4 | 44.8 | 309.7 |
| NA/c[a] | 57.0 | 353.8 | (26) | 24.6385 | 18 | 6.8 | 82.2 | 209.9 |
| NA/c[b] | 55.5 | 351.2 | (68) | 65.7014 | 29 | 3.2 | 81.5 | 192.6 |
| | | | *Elephant Head Quartz Monzonite[c] (31.7°N, 249.1°E)* | | | | | |
| EH1 | −61.5 | 156.0 | 4 | 3.9751 | 120 | 8.4 | 68.1 | 196.0 |
| EH2 | −43.3 | 163.2 | 4 | 3.9807 | 157 | 7.4 | 73.9 | 139.7 |
| EH3 | −48.1 | 163.2 | 6 | 5.9804 | 252 | 4.2 | 75.3 | 153.5 |
| EH4 | −46.4 | 167.4 | 7 | 6.9506 | 122 | 5.5 | 78.4 | 142.3 |
| EH5 | −41.5 | 171.6 | 6 | 5.9348 | 77 | 7.7 | 79.3 | 114.6 |
| Average | −48.2 | 164.9 | (5) | 4.9541 | 87 | 8.2 | 76.7 | 155.0 |
| | | | *Gringo Gulch Pluton[c] (31.6°N, 249.2°E)* | | | | | |
| GP1 | −57.9 | 145.7 | 6 | 5.9940 | 831 | 2.3 | 61.3 | 182.7 |
| GP2 | −57.7 | 159.6 | 6 | 5.9954 | 1092 | 2.0 | 72.0 | 187.0 |
| GP3 | −53.7 | 159.3 | 8 | 7.9687 | 224 | 3.7 | 72.4 | 173.4 |
| GP4 | −62.2 | 142.4 | 5 | 4.9410 | 68 | 9.4 | 58.3 | 192.0 |
| GP5 | −60.2 | 128.6 | 6 | 5.9943 | 876 | 2.3 | 48.2 | 187.3 |
| GP6 | −63.1 | 155.3 | 8 | 7.9898 | 685 | 2.1 | 66.8 | 200.2 |
| GP7 | −63.9 | 159.2 | 5 | 4.9892 | 370 | 4.0 | 68.7 | 206.2 |
| GP8 | −63.7 | 167.4 | 7 | 6.9930 | 858 | 2.1 | 73.1 | 217.5 |
| GP9 | −63.8 | 129.5 | 7 | 6.9825 | 343 | 3.3 | 48.9 | 193.8 |
| GP10 | −66.6 | 147.2 | 9 | 8.9613 | 207 | 3.6 | 59.9 | 204.2 |
| GP11 | −62.8 | 143.4 | 8 | 7.9825 | 399 | 2.8 | 62.3 | 195.3 |
| GP12 | −71.1 | 161.9 | 6 | 5.9895 | 474 | 3.1 | 62.8 | 226.7 |
| GP13 | −67.6 | 163.3 | 8 | 7.9893 | 651 | 2.2 | 67.4 | 220.8 |
| GP14 | −65.1 | 160.3 | 8 | 7.9871 | 542 | 2.4 | 68.4 | 210.6 |
| GP15 | −51.4 | 148.1 | 6 | 5.9910 | 557 | 2.8 | 63.0 | 168.8 |
| GP16 | −51.7 | 153.1 | 7 | 6.9453 | 110 | 5.8 | 67.3 | 168.4 |
| Average | −61.9 | 151.2 | (16) | 15.8620 | 109 | 3.6 | 64.6 | 194.2 |
| | | | *Gringo Gulch Volcanics[d] (31.5°N, 249.5°E)* | | | | | |
| Average | −58.8 | 167.5 | (25) | 24.9646 | 678 | 1.1 | 77.0 | 201.3 |
| | | | *Patagonia Granodiorite (31.2°N, 249.2°E)* | | | | | |
| PG1 | 41.3 | 355.1 | 6 | 5.7627 | 21 | 14.9 | 81.3 | 100.9 |
| PG2 | 54.4 | 334.9 | 8 | 7.6777 | 22 | 12.2 | 68.7 | 175.9 |
| PG3 | 48.3 | 343.7 | 8 | 7.8086 | 37 | 9.3 | 75.8 | 155.4 |
| PG4 | 52.5 | 343.9 | 7 | 6.7090 | 21 | 13.6 | 76.2 | 171.0 |
| | −47.0 | 334.7[e] | 11 | 10.4294 | 18 | 11.2 | 25.8 | 94.0 |
| PG5 | 47.3 | 329.0 | 2 | 1.9528 | 21 | 57.3 | 63.1 | 161.3 |
| | −72.4 | 27.9[e] | 7 | 6.8487 | 40 | 9.7 | 1.9 | 234.7 |
| PG6 | −62.3 | 14.4[e] | 9 | 8.4110 | 14 | 14.5 | 14.0 | 58.6 |
| Average | 49.1 | 341.8 | (5) | 4.9566 | 92 | 8.0 | 74.3 | 159.6 |
| | | | *Corral Canyon Sequence[f] (31.4°N, 249.3°E)* | | | | | |
| Average | 51.0 | 326.1 | (12) | 11.5323 | 24 | 9.1 | 61.2 | 172.6 |
| Average/c | 19.9 | 339.4 | (12) | 11.5963 | 32 | 7.8 | 61.8 | 116.0 |

$I$ and $D$ are inclination and declination of paleomagnetic directions uncorrected (/c, corrected) for structural tilt; $N$ is number of samples (sites) analyzed; $R$ is vector sum of $N$ unit vectors; $k$ is the concentration parameter [*Fisher*, 1953]; $\alpha_{95}$ is the radius of 95% confidence; $\lambda$ and $\phi$ are north latitude and east longitude of virtual geomagnetic pole.

[a]Expected Late Cretaceous North American (NA) direction for the Santa Rita Mountains, Arizona, from *Gunderson and Sheriff* [1991]. Statistical parameters are for pole.

[b]Expected Paleocene North American direction for the Santa Rita Mountains from *Diehl et al.* [1983, Table 6]. Statistical parameters are for pole.

[c]Unpublished data provided by R. F. Butler, Department of Geosciences, University of Arizona.

[d]From *Vugteveen et al.* [1981, Table 2].

[e]Not included in unit mean.

[f]From *May et al.* [1986, Table 3].

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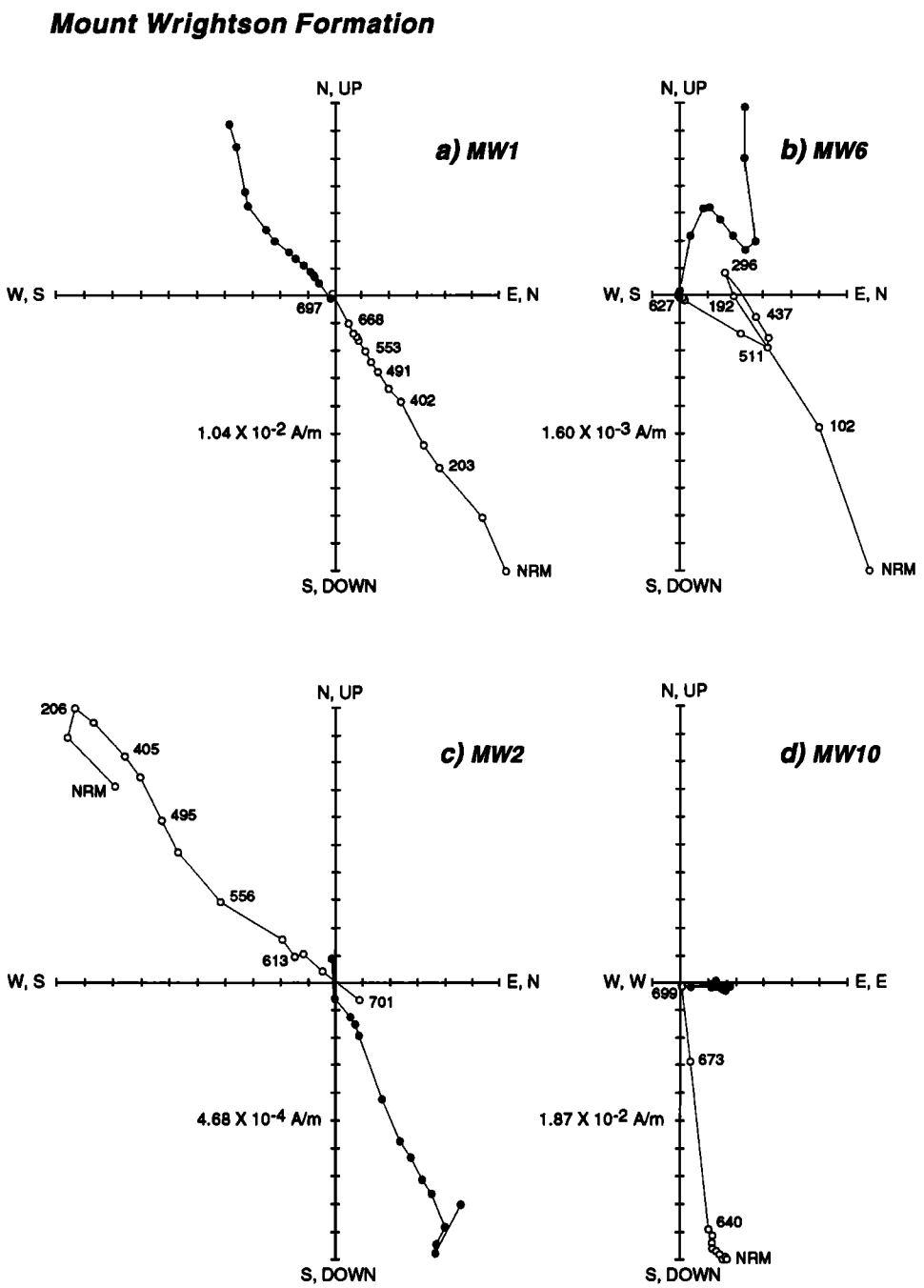



**Figure 3.** Orthogonal projections of thermal demagnetization data for select specimens of the Mount Wrightson Formation. Solid symbols indicate projection onto the horizontal plane, and open symbols indicate projection onto the vertical plane. (a) Dacitic rocks from site MW1 showing single high-temperature normal-polarity and low-temperature viscous components of magnetization. (b) Ash flow tuff from site MW6 showing two high-temperature normal and reversed polarity components and a low-temperature viscous component. (c) Ash flow tuff from site MW2 showing reversed polarity component and a poorly defined viscous component. (d) Ash flow tuff from site MW10 showing near-vertical in situ direction.

and the unit-mean direction for the Mount Wrightson Formation ($I = -47.3°$, $D = 153.7°$, $N = 9$, $\alpha_{95} = 9.1°$) is statistically indistinguishable (95% confidence level [*McFadden and Lowes*, 1981]) from that for the Elephant Head pluton ($I = -48.2°$, $D = 164.9°$, $N = 5$, $\alpha_{95} = 8.2°$). The structural correction for the Mount Wrightson Formation brings these eastward directions into general agreement with the expected North American direction, implying that the plutonic rocks were deformed along with the Mount Wrightson Formation, likely by Neogene Basin and Range faulting.

Paleomagnetic directions for ash flow tuffs of the Middle Jurassic Corral Canyon sequence to the south (Figure 2) are shown in Figure 6c. Also shown are site- and unit-mean directions for the Paleocene Patagonia Granodiorite. Primar-

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ily of normal-polarity, the Corral Canyon rocks also have a mean direction ($I$ = 51.0°, $D$ = 326.1°, $N$ = 12, $\alpha_{95}$ = 9.1°) that is indistinguishable from the Patagonia Granodiorite mean direction ($I$ = 49.1°, $D$ = 341.8°, $N$ = 5, $\alpha_{95}$ = 8.0°) at the 95% confidence level [*McFadden and Lowes*, 1981].

Samples from sites MW9, MW10, and MW11 show steep in situ characteristic magnetizations (Figure 7) that are different in direction from those for the other Mount Wrightson Formation sites. Sites MW9 (red tuff) and MW10 (gray tuff) are from the same ash flow tuff but are from outcrops that are different in color. After structural correction these directions are to the northeast ($I$ = 60.8°, $D$ = 56.1°, $N$ = 3, $\alpha_{95}$ = 5.4°) but have inclinations similar to that of the expected Late Cretaceous ($I$ = 57.0°, $D$ = 353.8°) and Paleocene ($I$ = 55.5°, $D$ = 351.2°) directions. These overprinted rocks may have been subjected to clockwise vertical axis rotation (~60°) with respect to stable North America before the rocks above them were remagnetized, similar to rotations observed in Upper Jurassic tuffs in the nearby Canelo Hills and Huachuca Mountains [*Hagstrum and Lipman*, 1991b]. These apparently rotated magnetizations, however, could also be explained by short-term magnetization acquisition during high-amplitude paleosecular variation of the geomagnetic field.

## Discussion

Similar paleomagnetic directions (in situ) for the Late Cretaceous Elephant Head pluton and the host Lower Jurassic Mount Wrightson Formation (Figure 6b) imply that the Mount Wrightson Formation was remagnetized by thermochemical effects associated with emplacement of younger plutons. Minor hematite is commonly a product of deuteric alteration within ash flow tuffs, but its presence as the dominant remanence carrier is often associated with areas of hydrothermal alteration, in particular, potassium metasomatism [*McIntosh*, 1991]. Multiple polarity directions in individual specimens, and between sampling sites, imply a lengthy period, or multiple episodes, of magnetization acquisition. Extensive mineralization in the region and a K-Ar age of ~67 Ma for altered rocks of the Mount Wrightson Formation [*Shafiqullah et al.*, 1980] also point to alteration and remagnetization within the Mount Wrightson Formation in Late Cretaceous and early Tertiary time. Preservation of dual polarity secondary magnetizations and varying magnetic carriers (magnetite and hematite) within ash flow tuffs of the sampled section, however, suggest that these rock sequences were not chemically or magnetically homogenized by hydrothermal fluid circulation.

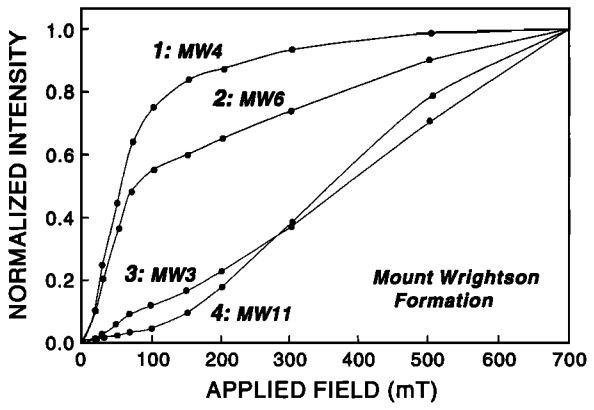



**Figure 4.** Acquisition of isothermal remanent magnetization (IRM) curves for specimens of the Mount Wrightson Formation. Behavior for specimens from (1) site MW4 indicates magnetite as the remanence carrier, (2) site MW6 indicating the presence of magnetite and hematite, (3) site MW3 (and MW1–2, MW7–8) indicating primarily hematite, and (4) site MW11 (and MW5, MW9–10) indicating hematite.

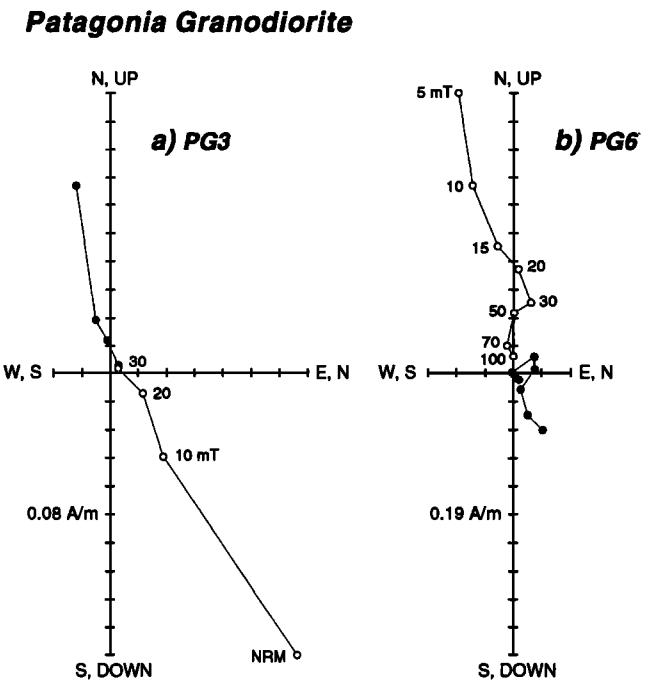



**Figure 5.** Orthogonal projections of alternating field demagnetization data for select specimens of the Patagonia Granodiorite. Solid symbols indicate projection onto the horizontal plane, and open symbols indicate projection onto the vertical plane. (a) Site PG3 specimen showing the characteristic normal polarity component of magnetization. (b) Site PG6 specimen showing a near-vertical reversed-polarity component.

Ash flow tuffs of the Middle Jurassic Corral Canyon sequence and Paleocene Patagonia Granodiorite in the Patagonia Mountains (Figure 2) also have similar characteristic paleomagnetic directions (Figure 6c), indicating that these Jurassic rocks also could have been remagnetized by thermochemical effects associated with emplacement of the younger granodiorite. In addition, the Corral Canyon sequence has similar geologic characteristics to the Mount Wrightson Formation: It is exposed along strike on the eastern flank of a mountain range cored by a Late Cretaceous to early Tertiary intrusion, the sampling locality is probably within this intrusion's hydrothermal aureole, and hematite is the dominant remanence carrier in the ash flow tuffs. *May et al.* [1986], however, argued that the Corral Canyon rocks retain primary remanence directions because (1) they show both normal and reversed polarities, (2) the angular standard deviation ($\sigma$) of the corrected virtual geomagnetic poles (VGPs) agrees well with expected values from global compilations of VGP dispersion versus latitude

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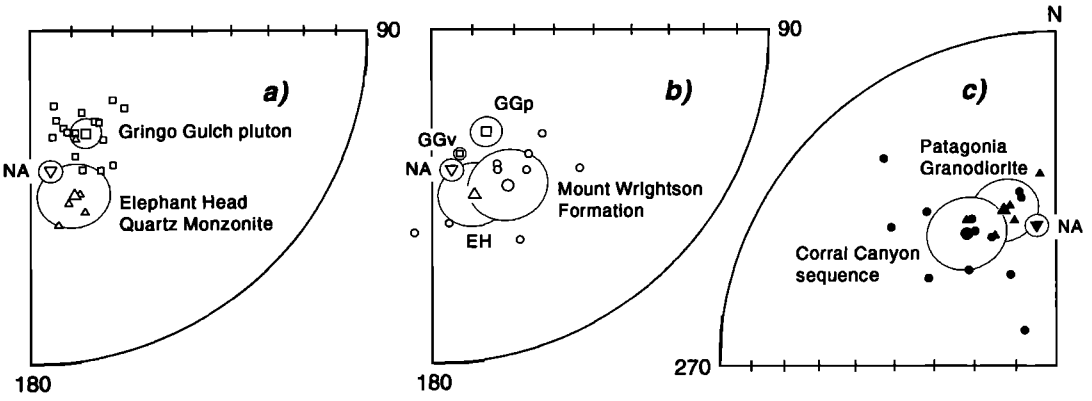



**Figure 6.** Equal-area plots of site and unit (with 95% confidence limits) mean directions (see Table 1) for (a) the Elephant Head Quartz Monzonite (EH, triangles) and Gringo Gulch pluton (GGp, squares), (b) the Mount Wrightson Formation including the Gringo Gulch Volcanics (GGv), and (c) the Corral Canyon sequence [*May et al.*, 1986] and Patagonia Granodiorite (triangles). NA (inverted triangle), expected North American Paleocene direction for southern Arizona [*Diehl et al.*, 1983]. Solid symbols indicate projection from the lower hemisphere, and open symbols indicate projection from the upper hemisphere.

for the last 5 m.y., (3) dispersion of the site-mean directions is reduced after structural correction, (4) the samples appear unmetamorphosed and have a 172 ± 6 Ma Rb-Sr whole rock age, and (5) the average paleomagnetic pole is similar to that for the Upper Jurassic Glance Conglomerate in the nearby Canelo Hills [*Kluth et al.*, 1982].

Paleomagnetic directions for the Corral Canyon rocks are mostly of normal polarity, consistent with directions for the Patagonia Granodiorite (Figure 5a and Table 1). Three sites in the Patagonia Granodiorite also have south directed, negative-inclination directions indicating that magnetization acquisition in these rocks spanned two polarity intervals (Figure 5b). Only one accepted site in the Corral Canyon sequence shows a reversed polarity direction, but two other sites have negative-inclination magnetizations interpreted as transitional field directions by *May et al.* [1986]. Similar to the Mount Wrightson Formation, the normal- and reversed-polarity directions in the Corral Canyon sequence may indicate that these rocks were episodically remagnetized by the Patagonia Granodiorite over a time period including both reversed- and normal-polarity geomagnetic fields. In addition, episodic remagnetization would likely result in random sampling of paleosecular variation and could explain why $\sigma$ is similar to that expected for a record of independent observations of the geomagnetic field.

The positive fold test (99% confidence using *McFadden*'s [1990] test (R. F. Butler, written communication, 1993)) for the Corral Canyon sequence implies that these rocks were magnetized prior to structural deformation, and *May et al.* [1986] suggest that the deformation was related to Late Cretaceous (Laramide) folding. It is also possible, however, that this deformation was simply differential crustal block tilting associated with Neogene Basin and Range faulting. The Mount Wrightson Formation to the north is situated along strike with the Corral Canyon sequence (Figure 2) and apparently was tilted to the northeast after remagnetization by the Elephant Head pluton. Similar directions for the Mount Wrightson Formation and Corral Canyon rocks might indicate that they were both tilted eastwards by Basin and Range faulting postremagnetization.

Although the Corral Canyon rocks have not been significantly metamorphosed since the Middle Jurassic, as indicated by a 172 ± 6 Ma Rb-Sr whole rock age [*May et al.*, 1986], they have been altered and probably lie within the eastern margin of the Patagonia Granodiorite's hydrothermal aureole. More altered and mineralized rocks occur along the American fault and other more permeable zones at the western edge of the Corral Canyon block. *May et al.* [1986] inferred that hematite within the welded tuffs of the Corral Canyon sequence resulted from high-temperature (deuteric) oxidation of magnetite either in the magma chamber or during eruption and emplacement. They also interpreted the similarity of magnetization directions for both magnetic phases to indicate early oxidation. *Hagstrum and Murchey*

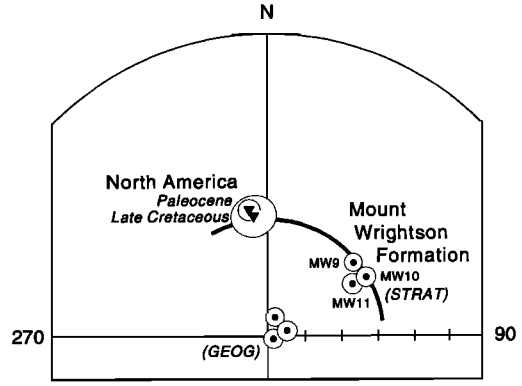



**Figure 7.** Equal-area plot of site-mean paleomagnetic directions for sites MW9–11 in the Mount Wrightson Formation (with 95% confidence limits) in both geographic (GEOG) and stratigraphic (STRAT) coordinates. Expected North American directions for southern Arizona based on paleomagnetic poles reported by *Gunderson and Sheriff* [1991] (Late Cretaceous) and *Diehl et al.* [1983] (Paleocene). Solid symbols indicate projection from the lower hemisphere. Small circle arc represents the expected Late Cretaceous inclination value.

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[1993], however, have found uniform secondary magnetizations in red chert (hematite) and basalt (magnetite) sequences in the Franciscan Complex of northern California, implying that low-temperature redox conditions governing the growth of secondary magnetic oxides can be lithologically dependent. These Franciscan cherts are also relatively unaltered and have not been isotopically (oxygen) homogenized by hydrothermal circulation [*Rumble and Spear*, 1983]. Thus low-temperature remagnetization of the Corral Canyon rocks in early Tertiary time would require an infiltrating pore fluid (probably noncirculating) which did not substantially modify the Rb-Sr (whole rock) isotopic systematics.

*May et al.* [1986] suggest that the similarity between poles for the Glance Conglomerate [*Kluth et al.*, 1982] and Corral Canyon sequence supports their interpretation of primary magnetizations within these rocks, but there are problems with the Glance Conglomerate pole as well. The Canelo Pass section is within the Parker Canyon caldera, and landslide megablocks within intracaldera tuff were misinterpreted by *Kluth et al.* [1982] as multiple volcanic units separated by sedimentary debris [*Lipman and Hagstrum*, 1992]. Moreover, these rocks have been affected by postemplacement potassium metasomatism [*Krebs and Ruiz*, 1987], and the remanent magnetization at five out of 15 sites in these rocks is carried by hematite [*Kluth et al.*, 1982].

*Kluth et al.* [1982] reported a Rb-Sr whole rock age of 148 ± 2 Ma [*Reynolds et al.*, 1986] for the Glance Conglomerate, but the isochon is based on samples from different ash flow sheets, and the one sample that defines the isochron's slope is an altered rock enriched in potassium [*Krebs and Ruiz*, 1987]. The Canelo Hills Volcanics near Canelo Pass also have a K-Ar biotite age of 177 ± 8 Ma [*Marvin et al.*, 1973]. Rb-Sr whole rock and K-Ar biotite ages for the Canelo Hills Volcanics, within the Parker Canyon caldera in the western Huachuca Mountains (Figure 2), range from 169 to 144 Ma [*Reynolds et al.*, 1986]. All these dates for the Canelo Hills Volcanics were obtained from altered rocks and might indicate the time of potassium metasomatism rather than the time of eruption and emplacement [*Krebs and Ruiz*, 1987]. The potassic alteration could have occurred shortly after emplacement or at a later time, perhaps postdeformation associated with caldera resurgence. A conglomerate test for the Glance Conglomerate is also based on samples from numerous rock types with magnetizations carried in both magnetite and hematite. Moreover, the dispersion might not be random [*Kluth et al.*, 1982], and the test becomes negative at the 95% confidence level if the reversed-polarity directions (secondary?) are inverted to normal-polarity directions.

Other poles supporting low-latitude APW and the J-2 cusp (Figure 1) are from the Upper Jurassic Morrison Formation [*Steiner and Helsley*, 1975] and the Middle Jurassic Summerville Formation [*Bazard and Butler*, 1992]. Paleomagnetic results for the Morrison Formation are complex, and results for only 100 samples out of 425 collected were used to calculate mean poles for these rocks [*Steiner and Helsley*, 1975]. Many samples were found to be contaminated by a recent field overprint, and means for selected samples of both polarities are not antipodal [*Steiner and Helsley*, 1975]. *Van Fossen and Kent* [1992b] suggest that the selected Morrison poles are biased to lower latitudes, and they point out that a pole based on less selected data for the lower part of the Morrison Formation is close to that for the upper part of the Morrison Formation; when combined, these data pass a reversal test at 95% confidence. Similarly, data for only 11 out of 35 sites (sedimentary horizons) in the underlying Summerville Formation passed reversal tests and were used to calculate a preferred paleopole at 56°N, 133°E [*Bazard and Butler*, 1992]. These rocks also have complex magnetizations, and *Bazard and Butler*'s [1992] pole is statistically distinct from that reported by *Steiner* [1978].

## Conclusions

The available evidence strongly supports remagnetization of the Lower Jurassic Mount Wrightson Formation by the adjacent Late Cretaceous Elephant Head and other intrusions. The secondary magnetization is carried by both magnetite and hematite, often as a single-component magnetization, and shows both normal- and reversed-polarity directions within a single sequence of ash flow tuffs. Such characteristics are usually cited as evidence of a primary magnetization. The Middle Jurassic Corral Canyon sequence has many of these same characteristics but also has paleomagnetic directions consistent with those of the nearby Paleocene Patagonia Granodiorite. The basic similarity of these Jurassic rocks and their paleomagnetic data with those of the Mount Wrightson Formation implies that they were also remagnetized during Late Cretaceous to early Tertiary plutonism.

The Rb-Sr whole rock age for the Corral Canyon sequence, however, indicates that the Rb-Sr whole rock system has remained closed since Jurassic time. The Corral Canyon rocks are near the margin of the Patagonia Granodiorite's hydrothermal aureole but show little evidence of significant alteration. Few studies are available on such "gray area" rocks between zones of marked alteration and fresh rock, so mechanisms by which these rocks can be remagnetized, while preserving a whole rock Rb-Sr system, are unclear. The Mount Wrightson Formation and basalt/chert sequences of the Franciscan Complex [*Hagstrum and Murchey*, 1993], although remagnetized, show that isotopic and chemical differences between adjacent layers can be preserved.

Problems also exist with the interpretation of paleomagnetic data for the Upper Jurassic Glance Conglomerate in the Canelo Hills (Figure 2). Again these rocks have characteristics similar to those of the Mount Wrightson Formation and Corral Canyon sequence. Moreover, they are situated within a caldera structure and have been potassium metasomatized; the reported conglomerate test may also be misleading. Thus the Jurassic volcanic rocks from southern Arizona which *May and Butler* [1986] used in constructing their APW path have complicated geologic histories, including possible remagnetization, and are apparently not as reliable as previously thought. This leaves only the pole for the Summerville Formation [*Bazard and Butler*, 1992] in support of low-latitude APW and the J-2 cusp. The remanent magnetizations of these continental red beds, however, are also complex, and interpretations of their primary magnetizations are not without uncertainties.

Jurassic paleopoles from other continents have been rotated into North American coordinates to test the configuration of the APW path. Middle Jurassic poles from stable parts of the Gondwana continents indicate North American

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APW near 75°N latitude [*Van Fossen and Kent*, 1990], whereas poles from Gondwanan orogenic belts imply Middle to Late Jurassic APW along the 70°N parallel (Figure 1) between the low-latitude Corral Canyon and high-latitude Moat Volcanics poles [*Van der Voo*, 1992]. These results and the evidence presented herein imply that the J-2 cusp and low-latitude configuration of North American APW, as proposed by *May and Butler* [1986], likely need revision and that clear documentation of North American Jurassic APW awaits a more thorough understanding of Jurassic rocks from North America as well as the Atlantic-bordering continents.

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